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Crosscutting Areas

No-Regret Bayesian Recommendation to Homogeneous Users

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Abstract. We introduce and study the online Bayesian recommendation problem for a recommender system platform. The platform has the privilege to privately observe a utility-relevant *state* of a product at each round and uses this information to make online recommendations to a stream of myopic users. This paradigm is common in a wide range of scenarios in the current internet economy. The platform commits to an online recommendation policy that utilizes its information advantage on the product state to persuade self-interested users to follow the recommendation. Because the platform does not know users' preferences or beliefs in advance, we study the platform's online learning problem of designing an adaptive recommendation policy to persuade users while gradually learning users' preferences and beliefs en route. Specifically, we aim to design online learning policies with no *Stackelberg regret* for the platform, that is, against the optimal benchmark policy in hindsight under the assumption that users will correspondingly adapt their responses to the benchmark policy. Our first result is an online policy that achieves double logarithmic regret dependence on the number of rounds. We also present an information-theoretic lower bound showing that no adaptive online policy can achieve regret with better dependency on the number of rounds. Finally, by formulating the platform's problem as optimizing a linear program with membership oracle access, we present our second online recommendation policy that achieves regret with polynomial dependence on the number of states but logarithmic dependence on the number of rounds.

Supplemental Material: All supplemental materials, including the code, data, and files required to reproduce the results, are available at https://doi.org/10.1287/opre.2023.0556.

Keywords: online learning • Bayesian recommendation • regret minimization • linear program

1. Introduction

Thanks to the rapid growth of modern technology, online platforms have become a major component of today's economy. By the end of 2021, at least 30 social platforms had more than 100 million active users, and 7 of them surpassed 1 billion users. According to a recent report by Knowledge Sourcing Intelligence LLP (2021), the global networking platforms market was valued at 192 billion U.S. dollars in 2019 and is projected to reach 940 billion U.S. dollars by 2026. This burgeoning economic sector raises numerous complex algorithmic challenges.

Within these platforms, recommended products or services are often tagged with labels signaling varying degrees of their relevance or quality to users. For example, music streaming platforms like Spotify use curated playlists with titles such as "Discover Weekly" and "Release Radar" to highlight the novelty or relevance of the recommended tracks. E-commerce platforms like Amazon use labels such as "Top Picks," "Best Sellers," and "Customers Also Bought" to categorize their product recommendations, each indicating the popularity level of the items. From the platforms' perspective, these labels act as informational signals that guide user behavior and decision making. However, in practice, the platform may not fully know users' preferences about the relevance or their prior beliefs about the quality of the recommended products. Below, we provide two concrete motivating applications.

Example 1 (Recommendation in Real-Time Short-Video Platforms). A prominent example of recommendations on social platforms, which serves as a major motivating application for this work, is real-time short-video recommendation systems, such as TikTok, Instagram Reels, and YouTube Shorts, as well as e-commerce platforms like Amazon Live and Taobao Live. Taking TikTok as an example, its signature feature is the "For You" feed, a stream of videos recommended to users in real time. In this application, a user repeatedly opens the app throughout the day and spends a significant

amount of time on the platform. The user aims to discover high-quality content, whereas the platform itself may have other objectives, such as maximizing user engagement or boosting content exposure: goals that may not always fully align with users' preferences. The platform evaluates videos based on quality metrics (e.g., through historical user feedback or other collected data) or characteristics like captions, sounds, and hashtags. Users, on the other hand, form prior beliefs about the quality of videos based on their past experiences with the platform or external information sources. Here, video quality refers to how engaging, interesting, or relevant users perceive the content to be. However, the platform often lacks knowledge of users' interests or prior beliefs, especially for new users. Despite this lack of knowledge, the platform must decide whether to recommend the video to the users. If the platform had perfect knowledge of users' preferences, it could reliably make recommendations that optimize its own objectives while keeping users satisfied (e.g., by balancing high-quality videos with less engaging ones). However, given the lack of such knowledge, the platform must learn to make such recommendations over time.

Example 2 (Recommendation in Dating Apps). Online dating apps (e.g., Tinder, Bumble, Coffee Meets Bagel) face similar challenges. Users on these platforms spend a significant amount of time every day on swiping and checking profiles, with the goal of building meaningful connections, whereas the platform itself may have additional goals, such as increasing revenue by giving extra visibility to paying subscribers, promoting new or underrepresented users to keep both sides of the market active, or favoring profiles that are predicted to generate high long-run engagement. These objectives can lead the platform to promote certain profiles for visibility even when they are not the user's top matches, so the platform's incentives are not perfectly aligned with the users' preferences. The platform can observe various characteristics of the suggested matches (e.g., detailed profiles and past interactions with other users). Each user, in turn, may have prior beliefs about the quality or compatibility of potential matches, based on their experiences with the app or external sources. However, the user's preference and prior belief are often unknown to the platform, particularly for new users or those with limited interactions. Despite this uncertainty, the platform must decide which profiles to recommend. If the platform had full knowledge of users' preferences and beliefs, it could consistently suggest matches that balance user satisfaction with its own goals. However, without such knowledge, the platform must gradually learn how to make better match suggestions over time.

Motivated by applications like the above, we introduce and study the *online Bayesian recommendation* problem, where a recommendation platform sequentially interacts with a homogeneous population of users, each sharing identical private preferences and beliefs. Here we describe the problem in the context of video recommendation. At each time, a video is displayed by the platform to an incoming user.[1] To capture the uncertain characteristics of the video, we study a *Bayesian* model. In this model, the payoff-relevant characteristics of a video are represented by a (random) *state* of the video. This state comprehensively encodes all critical payoff-relevant information, features, and characteristics of the video, such as content relevance, video quality, engagement metrics, and emotional impact. These elements are pivotal in influencing a user's experience and satisfaction when they choose to watch a video. The platform and user each have their own preferences over the video states, which are captured by their utility functions, respectively. We assume a natural *information asymmetry* between the platform and users—only the platform can privately observe the realized state of each video, whereas all users only have a prior belief about the video state. Notably, the platform also has its own prior belief over the video state, which may differ from the users' beliefs since they may form such beliefs from completely different sources. The platform designs a recommendation policy that makes different "levels" of recommendation based on its private information about the video, that is, its realized state. After observing the recommendation level, together with the user's initial belief, the user forms a posterior belief about the video and decides either to watch this video or skip it.

In the idealized situation where the platform knew both the users' preferences and prior beliefs, this sequential Bayesian recommendation problem reduces to a standard Bayesian persuasion problem and thus can be solved by linear programming (Kamenica and Gentzkow 2011, Alonso and Camara 2016, Dughmi and Xu 2019). This paper, however, addresses the more realistic yet challenging situation in which the platform does not know users' preferences nor users' prior beliefs. Therefore, the platform needs to adaptively update recommendation strategies based on the past users' behavior to maximize its own accumulated utility. To formalize the platform's goal, we use the *Stackelberg regret* to evaluate the performance of the platform's recommendation policy, and the goal of this paper is to design online policies to achieve no Stackelberg regret for the platform.[2]

1.1. Our Contributions and Techniques

In this work, we introduce a novel online Bayesian recommendation framework that addresses the challenge of making recommendations to arriving users with unknown yet homogeneous preferences, under

information asymmetry between the platform and users. Below, we overview our results and contributions for the case with only unknown user preferences but assuming users share the same common prior with the platform.[3]

When the platform has complete knowledge of users' preferences, the platform's problem can be viewed as a classic Bayesian persuasion problem with binary action (Kamenica and Gentzkow 2011). The optimal recommendation policies in all rounds are identical and can be solved independently as solving the optimal signaling scheme (that maps the realized state to a possibly random signal) in the Bayesian persuasion problem. The correspondence between the terminology in recommendation systems and in Bayesian persuasion is outlined in Table EC.1 in the Online Appendix. By the revelation principle, the optimal signaling scheme in hindsight is a *direct* signaling scheme with a binary recommendation level—that is, a binary signal where each signal corresponds directly to a specific action recommendation. In particular, the optimal signaling scheme associates each state with the users' *preference difference*, which represents how much the user prefers watching the video over not watching the video given this particular state. Then, the optimal signaling scheme specifies an *order* over all states based on the users' preference differences, as well as a *threshold state* so that it recommends every state above the threshold state in this order. The threshold state is selected to ensure that whenever the signaling scheme recommends a video, the user is indifferent between watching and skipping it.

When the platform lacks knowledge of users' preferences, it must use adaptive signaling schemes to learn the correct order of the states and the threshold state to achieve optimal long-term revenue. To understand and solve the platform's problem, we focus on two natural scenarios: (1) known ordinal preference—the order of the users' preference difference is known to the platform, but the exact differences are unknown[4]; and (2) unknown ordinal preference—the order of the users' preference difference is unknown to the platform. We summarize our results in Table 1.

Before diving into the detailed discussion of our results, we first highlight one crucial feature in our model—the feedback to the platform is *limited* and *probabilistic*. This feedback structure is a significant challenge that algorithms with low regret must overcome or bypass, setting our model apart from other classic models in the online learning literature. Specifically, in our model, a signaling scheme maps each video state to a possibly random signal; therefore, the platform only observes the user's response to this realized signal and not to other signals. The feedback is also probabilistic, as the realized signal depends on the realized state, which is drawn from an exogenous prior distribution. Consequently, the platform may incur substantial regret in order to learn the user's response to a specific signal realization or their preference for a specific state.

1.1.1. Known Ordinal Preference. For the known ordinal preference scenario, it suffices for the platform to learn which state is the threshold state in the optimal signaling scheme and how to recommend when this threshold state is realized. For this scenario, we show that there exists a conservative recommendation policy (`ConRP`; Algorithm 2), which achieves the double-logarithmic $O(\log\log T)$ regret, as stated in our first main result below (see Theorem 1 for the formal result).

Theorem (Informal). `ConRP` *achieves* $O(\log\log T)$ *regret.*

The key intuition behind the `ConRP` is an "unbalanced" utility structure, in which the platform has zero payoff under overoptimistic recommendations (e.g., a predicted high-user-value video ends up not being watched) but only slightly suboptimal payoff under overpessimistic recommendations. This "unbalanced" utility structure shares similarity to the seller's payoff (as a function of the charged price) in a classic dynamic pricing problem with unknown yet fixed buyer valuation (Kleinberg and Leighton 2003). Inspired by that literature, algorithm `ConRP` uses a "conservative" binary search to identify the threshold state and determine how to recommend on this threshold state. What sets our method apart from previous studies are the additional careful treatments to handle the aforementioned challenge on the limited and probabilistic feedback structure in our problem.

Table 1. Regret Bounds of the Online Bayesian Recommendation Problem

Settings		Upper bound	Lower bound
Known ordinal preference		$O(\log\log T)$[a] (Theorem 1)	$\Omega(\log\log T)$[a] (Theorem 4)
Unknown ordinal preference	Affine preference[b]	$O(\log\log T)$[a] (Proposition 1)	
	Arbitrary preference	$O(m2^{m-1}\log\log T \wedge m^6 \log^{O(1)}(mT))$ (Proposition 2, Theorem 2)	

Note. Here, m is the number of states, and T is the time horizon.
[a]These bounds do not depend on the number of states m.
[b]The user's utility is an affine function over the state.

1.1.2. Unknown Ordinal Preference. For the unknown ordinal preference scenario, the order and the threshold state specified in the optimal signaling scheme remain unknown. Because of the limited and probabilistic feedback feature, designing an online policy to pin down this order with logarithmic regret may appear impossible at first glance. However, we show that when the users' preferences are affinely dependent across the states (in this case, the user's expected utility can be uniquely determined by the mean of the user's posterior belief), one can still achieve regret with double logarithmic dependence on the number of rounds T, and moreover, this regret is independent of the number of states and also holds even for a continuous state space. On the other hand, for arbitrary users' preferences, a modified `ConRP`, which enumerates over all possible orders and prunes out the bad orders in the process, can lead to regret with still double logarithmic dependence on the number of rounds T but with an exponential dependence on the number of states m. The informal statement of these results is as follows (see Proposition 1 and Proposition 2 for the formal results).

Proposition (Informal). *For affine preferences, a modified* `ConRP` *has expected regret* $O(\log\log T)$*; for arbitrary preferences, a modified* `ConRP` *has expected regret* $O(m2^{m-1} \cdot \log\log T)$.

A caveat of the above results is that the regret exponentially depends on the number of states m. For a wide range of applications, it is reasonable to focus on problem instances with small m.[5] To shed light on instances with large m, we propose another policy, a linear program–based recommendation policy (`LP-RP`; Algorithm 5 in the Online Appendix), whose regret polynomially depends on m but logarithmically on T. We design `LP-RP` by formulating the problem as optimizing a linear program with membership oracle access. Specifically, the unique form of user feedback in our setting can be exploited to construct a membership oracle for the platform's linear program problem.

Theorem (Informal). `LP-RP` *achieves* $O(\text{poly}(m\log T))$ *regret.*

We note that similar ideas of formulating learning problems as optimizing linear programs have also been applied to other online learning problems such as contextual dynamic pricing (Leme and Schneider 2018) and the security game (Blum et al. 2014). However, our `LP-RP` requires additional special treatment to overcome the issue of probabilistic signals. Moreover, compared with using the separation oracle as in Leme and Schneider (2018) and Blum et al. (2014), our problem of linear optimization with membership oracle access is considerably harder. For instance, one key technical hurdle, which does not appear in previous works but our `LP-RP` has to overcome, is to construct an *interior point* inside the feasible region.

1.1.3. Lower Bound. Similar to the optimal signaling scheme, `ConRP` and `LP-RP` only use direct signaling schemes with *binary* signals. Such direct signaling schemes are prevalent in many real-world applications such as "For You" in TikTok. However, when the platform does not know and has to learn users' preferences, the revelation principle does not necessarily hold—that is, it is unclear whether restricting to direct signaling schemes with binary signals is still without loss of generality during learning. Our third main result provides an affirmative answer, showing that introducing more signals cannot improve the regret dependence on T for all scenarios we mentioned before.

Theorem (Informal). *No online policy can achieve a regret better than* $\Omega(\log\log T)$ *even for problem instances with binary state.*

1.1.4. Simulations. In Section EC.8 of the Online Appendix, we also provide numerical experiments to evaluate the empirical performance of our proposed algorithm and highlight some of its salient features. In particular, we evaluate the performance of our proposed algorithm `ConRP` and compare its performance with several benchmarks, including the benchmarks that use simple searching strategies to find out the optimal signaling scheme without considering the unique structure of our problem and the benchmarks that use simple signaling schemes. We observe that our algorithm significantly outperforms all of these benchmarks. The results not only demonstrate the benefits of using partial information revealing in the platform's problem but also show the efficiency of our algorithm.

1.2. Related Work

Our work connects to several strands of existing literature. First, when the user's preference is known and they share the same prior belief with the platform, the one-shot instantiation of our problem exactly follows the formulation of the canonical Bayesian persuasion problem (Kamenica and Gentzkow 2011). Bayesian persuasion concerns the problem where an informed sender (i.e., platform) designs an information structure (i.e., signaling scheme) to influence the behavior of a receiver (i.e., user). There is a growing literature, including our work, on studying the relaxation of one fundamental assumption in the Bayesian persuasion model: The sender perfectly knows the receiver's preference and their prior belief. There are generally two approaches to deal with the uncertainty from the sender's perspective: the robust approach (Kosterina 2018, Babichenko et al. 2021, Hu and Weng 2021,

Dworczak and Pavan 2022) that tries to design signaling schemes that perform robustly well for all possible receiver preferences, and the online learning approach (Castiglioni et al. 2020, 2021; Zu et al. 2021) that studies the regret minimization when the sender repeatedly interacts with receivers.[6] Our work falls into the second approach. In particular, Castiglioni et al. (2020) address the sender interacting with receivers who have an unknown type. They provide an algorithm with a regret guarantee $O(T^{4/5})$ but it has exponential running-time over the number of states. Zu et al. (2021) study a setting where the sender has an unknown prior distribution, and they require the sender to make incentive compatible (IC) signaling schemes at each round. They provide an algorithm with an $O(\sqrt{T})$ regret bound and demonstrate that it is tight whenever the receiver has five (or more) actions. Our work differs from the above works in many ways. First, instead of assuming unknown types, our setting directly relaxes the knowledge of the user's utilities. Second, we do not require the platform's signaling scheme to be IC at each round. Third, we achieve logarithmic regrets over the time horizon, and this is possible due to the special structure of the Bayesian recommendation problem.

Second, our work also relates to research on Bayesian exploration in multiarmed bandit (Kremer et al. 2014; Mansour et al. 2015, 2021). In both our Bayesian recommendation and their Bayesian exploration, the platform utilizes its information advantage to persuade the user to take the desired action, and the user observes the platform's message and forms their posterior that will be used to pick their optimal action. However, in Bayesian exploration, the platform is learning the true state of nature, which is realized at the very beginning and never changes afterward, and is required to make incentive-compatible action recommendation at each round. However, in our setup, the platform is learning the users' preferences and beliefs, and the state is realized independently across the time horizon. Additionally, our problem does not require the platform to make incentive-compatible recommendations. Thus, the analysis and the technique of this work are quite different from theirs.

Third, our setting with homogeneous users shares similarities with the fixed valuation in (contextual) dynamic pricing literature (Kleinberg and Leighton 2003, Leme and Schneider 2018, Lobel et al. 2018, Liu et al. 2021), where the logarithmic regrets are also achievable. Part of our analysis is also related to this line of literature. In particular, we prove our lower bound via a nontrivial reduction to the single-item dynamic pricing problem. Although it seems that our problem for multiple states shares similarities with the contextual dynamic pricing (e.g., we both need to learn an unknown vector: in our setting, it is the user's preference for product states, and in contextual pricing, it is the buyer's preference for product features), we note that there are significant differences in our problem structure such as the platform's actions and the probabilistic feedback from user s (see Section EC.2 in the Online Appendix for the detailed comparisons).

2. Preliminary

Motivated by the applications of short-video platforms, this paper introduces and studies the *Bayesian recommendation* problem.

2.1. Basic Setup

We begin by describing a static model and then introduce the online setup studied in this work.

In the static model, there are two players: a platform and a user. The platform wants to recommend a video to the user. The video is associated with a private state θ drawn from a finite set $[m] \triangleq \{1, \ldots, m\}$ according to a prior distribution $\lambda \in \Delta([m])$, which is common knowledge among both players. We use the notation θ to denote the state as a random variable and $i, j \in [m]$ as its possible realizations. The user has a binary-action set $\mathcal{A} = \{0, 1\}$ (i.e., not watch or watch) and a utility function $\rho : [m] \times \mathcal{A} \to \mathbb{R}$ mapping from the state of the video and the user's action to the user's utility. The platform has the utility function $\xi : [m] \times \mathcal{A} \to \mathbb{R}$ that depends on the user's action and the realized state. In addition, the platform has an information advantage in the sense that the platform can observe the realized video state but the user cannot. This advantage enables the platform to recommend the video in different levels based on the realized state. In particular, the platform can design a finite[7] signal space Σ where each signal $\sigma \in \Sigma$ represents a recommendation level for the video, and a signaling scheme $\pi = \{\pi(i, \cdot)\}_{i \in [m]}$, where $\pi(i, \cdot) \in \Delta(\Sigma), i \in [m]$ denotes the conditional probability distribution over the signals when the state i is realized, and we use $\pi(i, \sigma)$ as the conditional probability of sending signal $\sigma \in \Sigma$ at state i.

For ease of presentation, our main context will focus on a stylized setup where (i) the platform and the user share the same prior belief λ over $[m]$, and (ii) the platform has state-independent utility function and only benefits from the user's action 1 (i.e., watch/click), namely, $\xi(\theta, a) \equiv \xi(a)$, $\forall \theta \in [m]$ and $\xi(1) \geq \xi(0)$. In Section 7, we illustrate how our algorithms and results can be easily extended to general settings where (i) users might have different unknown prior beliefs and (ii) the platform has arbitrary utility functions. Without loss of generality, we normalize $\xi(a) = a$. Namely, the platform gains one unit of revenue if the user watches the video.

In this work, we consider the following repeated interaction between the platform and a population of users. All users share the same utility function $\rho(\cdot, \cdot)$,

which is *unknown* to the platform. The setting proceeds for T rounds. For each round, $t = 1, \ldots, T$:

1. The platform commits to a signaling scheme π_t with a signal space Σ_t.
2. A video with state $\theta_t \sim \lambda$ is realized (only observed by the platform), and a signal $\sigma_t \sim \pi_t(\theta_t, \cdot)$ is realized (observed by both players) subsequently according to designed signaling scheme π_t.
3. Upon seeing the signal σ_t, together with the knowledge of π_t, the user t updates the user's belief regarding the underlying state to a *posterior distribution*, a.k.a. the posterior, denoted by $\mu_t(\sigma_t, \cdot) \in \Delta([m])$, where $\mu_t(\sigma_t, i) \triangleq \frac{\lambda(i)\pi_t(i,\sigma_t)}{\sum_{j\in[m]}\lambda(j)\pi_t(j,\sigma_t)}$ is the posterior probability on state i.
4. With the posterior $\mu_t(\sigma_t, \cdot)$, the user t chooses an action $a_t(\sigma_t)$ that maximizes the user's expected utility, that is, $a_t(\sigma_t) = \arg\max_{a\in\mathcal{A}} \mathbb{E}_{\theta\sim\mu_t(\sigma_t,\cdot)}[\rho(\theta, a)]$.
5. The platform then derives the utility a_t.

Given a signaling scheme π, let $U(\pi)$ denote the platform's one-shot expected payoff,

$$U(\pi) = \sum_{i\in[m]} \sum_{\sigma\in\Sigma} \lambda(i)\pi(i,\sigma) \cdot a(\sigma),$$

where Σ is the associated signal space, and $a(\sigma) \in \mathcal{A}$ is the agent's best response action upon seeing signal σ from the signaling scheme π. The goal of the platform is to design an online policy that constructs signaling schemes $\{\pi_t\}_{t\in[T]}$ to maximize its long-term expected utility $\sum_{t\in[T]} U(\pi_t)$. We conclude this section with the following two remarks.

Remark 1 (Bayesian-Rational Behavior). Upon seeing a signal realization, the user is able to form the user's Bayesian posterior belief and then make the user's decision by maximizing the expected utility based on the current belief. This Bayesian-rational behavior follows the common assumption adopted in Bayesian persuasion literature and other literature that studies signaling as a way to reveal product quality/characteristic information.[8]

Remark 2 (Commitment Power). In the above interaction, the platform is assumed to have the commitment power, and the designed signaling scheme is known to the user. In practice, the platform might lack this commitment power and may not be able to change its signaling scheme daily. However, we note that users may engage with the platform over a specific duration (i.e., a time cycle), for example, staying on the platform for a while or returning to the platform several times. Throughout this duration, the platform may stick to the same signaling scheme, allowing the user to discern the underlying signaling scheme, as well as the user's best response to it. Thus, a single round in the above theoretical model can be equated to one practical time cycle.

2.2. Stackelberg Regret and Benchmark

We evaluate the performance of an online policy by its Stackelberg regret (Chen et al. 2020) against the *optimal policy in hindsight*. The optimal policy in hindsight knows user s' utility function $\rho(\cdot,\cdot)$, and maximizes the platform's long-term expected utility. Because users are all identical, the optimal policy in hindsight commits to the same signaling scheme π^* (see its characterization in program $\mathcal{P}^{\mathrm{opt}}$ and Lemma 1) for every round $t \in [T]$.

Definition 1. Given the user's utility function ρ, let π^* be the optimal signaling scheme. The Stackelberg regret of online policy ALG is

$$\mathbf{REG}[\mathtt{ALG}] \triangleq \sum_{t\in[T]} U(\pi^*) - \mathbb{E}_{\pi_1,\ldots,\pi_T}\left[\sum_{t\in[T]} U(\pi_t)\right],$$

where π_t is the signaling scheme committed by ALG in each round $t \in [T]$.

Different from the regret notation (e.g., external regret) in classic single-agent no-regret learning literature (Blum and Mansour 2007), the Stackelberg regret compares to the optimal policy in hindsight, where user s have the opportunity to regenerate a different history by best responding to the new signaling scheme π^*. In the remainder of the paper, we simplify the terminology Stackelberg regret as regret.

To facilitate our analysis, we introduce one auxiliary variable, $\delta(i) \triangleq \rho(i,1) - \rho(i,0)$, to represent how much the user prefers action 1 over action 0 given state i. To make the problem nontrivial, we make the following two assumptions on the user's utility function throughout this paper.

Assumption 1. *There exists at least one state $i \in [m]$ such that $\delta(i)\lambda(i) > 0$.*

Assumption 2. *The user's utility function satisfies that $\sum_{i\in[m]} \delta(i)\lambda(i) < 0$.*

Assumption 1 ensures that there is at least one state where the user prefers action 1 over action 0. Without this, the problem becomes trivial, as the user would always choose action 0 regardless of the signaling scheme, and any online policy would achieve zero regret. Assumption 2 ensures that, under a no-information-revealing signaling scheme, the user's best response (i.e., the prior-best action) is to choose action 0. Without this, the platform could commit to a no-information-revealing scheme to achieve optimal utility, as the user would always choose action 1.

When the users' utility function $\rho(\cdot,\cdot)$ is known to the platform, the optimal signaling scheme in hindsight can be characterized as follows. By the revelation principle (Kamenica and Gentzkow 2011), the optimal signaling scheme is a direct signaling scheme with binary

signal space $\Sigma = \{0,1\} \equiv \mathcal{A}$ that each signal corresponds to an action recommendation. In particular, it can be solved by a linear program as follows:

$$\begin{aligned} \pi^* = \arg\max_{\pi} \quad & \sum_{i\in[m]} \lambda(i)\pi(i,1) && \text{s.t.} \\ \text{(IC)} \quad & \sum_{i\in[m]} \delta(i)\lambda(i)\pi(i,1) \geq 0 && \\ & \pi(i,1) + \pi(i,0) = 1 && i \in [m] \\ & \pi(i,1) \geq 0, \pi(i,0) \geq 0 && i \in [m]. \end{aligned} \qquad (\mathcal{P}^{\text{opt}})$$

The objective in the above program $U(\pi) = \sum_{i\in[m]} \lambda(i)\pi(i,1)$ is the platform's expected utility from an incentive compatible, direct signaling scheme π. The first constraint is an IC constraint to ensure that it is the user's best response to follow the recommended action realized from the signaling scheme; that is, taking action 1 is indeed the user's optimal action given the user's posterior when action 1 is recommended. We note that in Program ($\mathcal{P}^{\text{opt}}$), the IC constraint for action 0 is omitted because the IC constraint for action 1 and Assumption 2 can jointly imply the IC constraint for action 0. To see this, notice that the IC constraint for action 0 is $\sum_{i\in[m]} \delta(i)\lambda(i)\pi(i,0) < 0$. Rearranging the IC constraint for action 1, we obtain $\sum_{i\in[m]} \delta(i)\lambda(i)\pi(i,0) \leq \sum_{i\in[m]} \delta(i)\lambda(i)$, where the right-hand side is weakly smaller than zero due to Assumption 2.

For ease of presentation, with slight abuse of notation, we use $\pi^*(i) \triangleq \pi^*(i,1)$ and thus $\pi^*(i,0) \equiv 1 - \pi^*(i)$. Program $\mathcal{P}^{\text{opt}}$ can be interpreted as a fractional knapsack problem, where the budget is zero, and each state i corresponds to an item with value $\lambda(i)$ and (possibly negative) cost $\delta(i)\lambda(i)$. Thus, its optimal solution π^* has the following characterization.

Lemma 1 (For Example, Renault et al. 2017). *The optimal signaling scheme π^* in hindsight is the optimal solution of linear Program ($\mathcal{P}^{\text{opt}}$). There exists a threshold state $i^\dagger \in [m]$ such that* (a) *for every state $i \neq i^\dagger$, $\pi^*(i) = \mathbb{1}[\delta(i) \geq \delta(i^\dagger)]$, and* (b) $\pi^*(i^\dagger) = -\frac{\sum_{i\neq i^\dagger} \delta(i)\lambda(i)\pi^*(i)}{\delta(i^\dagger)\lambda(i^\dagger)}$.

In words, Lemma 1 states that the signaling scheme π^* reveals whether the state is above or below[9] a threshold state $i^\dagger$, with possible randomization at state $i^\dagger$.

The above Lemma 1 highlights the significance of both the cardinal value and the order of user preference differences $\{\delta(i)\}$ to characterize the optimal signaling scheme in hindsight. As we mentioned earlier, the user preference $\rho(\cdot,\cdot)$ is unknown to the platform. Consequently, the platform is unaware of both the cardinal value and the order of $\{\delta(i)\}$ and thus needs to learn these quantities by adaptively changing its signaling schemes. To address this intricate learning challenge, we initially study the scenario where only the cardinal value of $\{\delta(i)\}$ is unknown to the platform, whereas its order is known to the platform (see Section 3). Building on the insights from this scenario, we then discuss the more complex situation where the platform lacks knowledge of both the cardinal value and the order of $\{\delta(i)\}$ (see Sections 4 and 5).

2.3. Useful Subroutine for Checking the Incentive Compatibility

When the user's utility function $\rho(\cdot,\cdot)$ is unknown, the standard revelation principle fails. As a consequence, restricting to binary signal space (e.g., {"recommended", "not recommended"}) is *not* without loss of generality. Nonetheless, as we formally show later, restricting to the subclass of signaling schemes with binary signal space does not hurt the optimal regret. We now formally define such signaling schemes as follows.

Definition 2. A *direct signaling scheme* π is a mapping from states into probability distributions over actions recommended to the user.

With slight abuse of notation, for every direct signaling scheme π, we use $\pi(i) \triangleq \pi(i,1)$ and thus $\pi(i,0) \equiv 1 - \pi(i)$. When facing a direct signaling scheme π, the user takes the action that maximizes the user's expected utility given the user's posterior. We say a signaling scheme π is IC if the user takes action 1 as long as action 1 is recommended by the signaling scheme π. The proofs of Lemma 2, Lemma 3, and Lemma 4 are deferred to Section EC.3 in the Online Appendix.

Lemma 2. *A direct signaling scheme π is* IC *if and only if* $\sum_{i\in[m]} \delta(i)\lambda(i)\pi(i) \geq 0$.

Before we finish the preliminary section, we provide Procedure 1 as a useful subroutine that will be used in our online policies. Procedure 1 takes a direct signaling scheme as input and determines whether this direct signaling scheme is IC. Its correctness guarantee is given in Lemma 3, and the regret guarantee incurred by implementing Procedure 1 is given in Lemma 4.

Procedure 1 (`CheckIC`(π))

Input: a direct signaling scheme π

Output: True/False–whether π is IC; or `round-exhausted` if there is no round left

while *there are rounds remaining* **do**
  `/* suppose now is round` t `*/`
  Commit to signaling scheme π toward user t.
  if $\sigma_t = 1$ and $a_t = 1$, **then** return True
  else if $\sigma_t = 1$ and $a_t = 0$, **then** return False
  else if $\sigma_t = 0$ and $a_t = 1$, **then** return False
  else move to next round, that is, $t \leftarrow t + 1$
end
return `round-exhausted`

Lemma 3. *Given a direct signaling scheme π, Procedure 1 returns* true *only if π is* IC *and returns* false *only if π is not* IC.

We note that Procedure 1 does not include the case for $\sigma_t = 0$ and $a_t = 0$ because when this case happens, the user's response (i.e., taking $a_t = 0$) does not convey any information about whether the signaling scheme is IC or not. In particular, when this case happens, we know that $\sum_{i\in[m]} \lambda(i)\delta(i)(1-\pi(i)) < 0$, and this inequality does not imply the IC inequality $\sum_{i\in[m]} \delta(i)\lambda(i)\pi(i) \geq 0$ defined as in Lemma 2 due to $\sum_{i\in[m]} \lambda(i)\delta(i) < 0$ in Assumption 2. Thus, Procedure 1 stops either when it encounters one of the listed three cases (in which the user's response can be used to infer whether the committed signaling scheme is IC or not) or when the time rounds are exhausted. As long as Procedure 1 returns true\false, the platform knows for sure whether the signaling scheme is IC or not.

Lemma 4. *Given a direct signaling scheme π, the expected regret of Procedure 1 is at most $\frac{U(\pi^*)}{\sum_i \lambda(i)\pi(i)} - \mathbb{1}[\pi \text{ is IC}]$.*

The intuition behind Lemma 4 is as follows. Given a direct signaling scheme π, as long as its probability (i.e., the value $\sum_i \lambda(i)\pi(i)$) for recommending action 1 is a constant approximation to the optimal payoff $U(\pi^*)$, then the expected regret of Procedure 1 for checking its incentive compatibility can be upper bounded by this constant. However, if this probability is small compared with $U(\pi^*)$, then the expected incurred regret can be very large, regardless of the incentive compatibility of π or the value of $U(\pi^*)$.

3. Conservative Recommendation Policy for Known Ordinal Preference

In this section, we provide our first result, conservative recommendation policy (ConRP), which is an online policy that can achieve low regret when the order of the user's preference differences $\{\delta(i)\}$ is known to the platform. We upperbound its regret by $O(\log\log T)$. Combining with the lower bound $\Omega(\log\log T)$ of regret for any online policy developed in Section 6, our policy ConRP is regret-optimal. The results of this section will be served as a building block for the algorithm design when the order of $\{\delta(i)\}$ is not known in advance.

Before diving into our results, let us highlight one of the main challenges in the design of a good online policy. In our problem, the platform's feedback is *limited* and *probabilistic*. Specifically, when signaling scheme π_t is used in round t and signal $\sigma_t \sim \pi_t(\theta_t)$ is realized, the platform only observes the user's action under signal σ_t and learns its corresponding payoff, but nothing about its payoff under other signals. Meanwhile, this feedback is also probabilistic because the realized signal σ_t depends on the realized state θ_t. Because of these two features of the feedback, some natural tasks toward learning the user's utility may not be completed easily. Here are two illustrative examples.

3.0.1. Identifying the Signs of $\{\delta(i)\}$. Recall that the optimal signaling scheme in hindsight π^* follows from a threshold signaling scheme; it recommends action 1 deterministically for all states above a threshold state $i^\dagger$, recommends action 1 randomly at threshold state, and recommends action 0 deterministically for all states below $i^\dagger$. Following the same logic, a natural attempt to design a good online policy is to try to identify the threshold state and the states that are above the threshold state. However, it is unclear how to identify the threshold state. In fact, it is even challenging to identify the sign of $\delta(i)$ for a state i. To see this, ideally, identifying the sign of $\delta(i)$ needs to solicit the user's action when the user's posterior belief is concentrated on state i when a particular signal is realized. A signaling scheme π with $\pi(j) = \mathbb{1}[j = i]$, $\forall j \in [m]$ can shape the user's posterior belief to be concentrated on state i when a signal 1 is realized. However, because the feedback is limited, such a signaling scheme cannot collect useful information whenever other signal is realized. Consequently, it bears a large regret if it happens to be the case when $\lambda(i)$ is small.

3.0.2. Determining If $U(\pi^*) \geq C$. Consider a problem instance with $m = 2$ states. Suppose the platform knows that $\delta(1) > 0$, and $\delta(2) < 0$. This implies that the threshold state $i^\dagger = 2$, and state 1 is above threshold state 2. Note that a good online policy should be able to approximately identify the value of $U(\pi^*)$. Now, suppose the platform only wants to determine whether $U(\pi^*) \geq C$. If the platform can determine whether following this natural signaling scheme $\pi(1) = 1$ and $\pi(2) = (C - \lambda(1))/\lambda(2)$, is IC or not, then the platform can determine whether $U(\pi^*) \geq C$.[10] However, because the feedback is probabilistic, it takes $1/C$ rounds (in expectation) to learn the incentive compatibility of π. Thus, even if $U(\pi^*)$ is small (i.e., $U(\pi^*) = o(1)$), by Lemma 4, the aforementioned attempt bears a superconstant regret as long as $C = o(U(\pi^*))$.

3.1. Toward $O(\log\log T)$ Regret

Despite the aforementioned challenges, in this section, we present ConRP that can have $O(\log\log T)$ regret guarantee. Our algorithm is inspired by Kleinberg and Leighton (2003) yet requires necessary and nontrivial modifications to account for the unique feedback structure in our problem. In Section EC.2 in the Online Appendix, we further discuss the connection between

our model with the dynamic pricing problem and how our new algorithmic ingredients can be used to solve a new variant of the dynamic pricing problem.

3.1.1. Overview of the Algorithm. In the ConRP, the whole T rounds are divided into the *exploring* phase and the *exploiting* phase. The exploring phase has two subphases. The first subphase (i.e., *exploring phase I*) identifies a lower bound and an upper bound of $U(\pi^*)$; that is, it identifies an IC signaling scheme $\underline{\pi}$ with $\underline{U} := U(\underline{\pi})$ such that $\underline{U} \le U(\pi^*) \le 2\underline{U}$. Note that once we narrow down the value of $U(\pi^*)$ to be in the interval $[\underline{U}, 2\underline{U}]$, with the IC signaling scheme $\underline{\pi}$, we can ensure that expected regret to check the incentive compatibility of the signaling schemes in the later rounds is at most a constant, which addresses the second challenge (i.e., determining if $U(\pi^*) \ge C$) we just mentioned before. We will show that the expected cumulative regret in exploring phase I is $O(1)$. The second subphase (i.e., *exploring phase II*) identifies a signaling scheme $\pi^\dagger$ whose per-round expected regret is $1/T$, and we will show that its expected cumulative regret is at most $O(\log\log T)$. The identified signaling scheme $\pi^\dagger$ from the exploring phase II is used in the remaining rounds that are considered the exploiting phase that induces $O(1)$ expected cumulative regret. See Algorithm 2 for a formal description.

3.1.2. Subclass of Direct Signaling Schemes $\{\pi^{(u)}\}$. Our online policy will repeatedly consider a subclass of direct signaling schemes. Recall Program ($\mathcal{P}^{\mathrm{opt}}$) indicates that the optimal signaling scheme in hindsight π^* can be thought of as the optimal solution of a fractional knapsack problem, where each state i corresponds to an item with value $\lambda(i)$ and cost $\omega(i)$. This observation implies that there must exist a total order r^* over all states with respect to their true bang-per-buck $\delta(i) = \lambda(i)/\omega(i)$. Given an arbitrary number $u \in [0,1]$, we define $\pi^{(u)}$ to be the direct signaling scheme as follows: There exists a threshold state $i^\dagger$ such that (a) for every state $i \ne i^\dagger$, $\pi^{(u)}(i) = \mathbb{1}[\delta(i) > \delta(i^\dagger)]$ and (b) $\pi^{(u)}(i^\dagger) = \frac{u - \sum_{i \ne i^\dagger} \lambda(i)\pi^{(u)}(i)}{\lambda(i^\dagger)}$ (recall that because the platform knows the order of user's preference difference $\{\delta(i)\}$, this signaling scheme is well defined). As a sanity check, observe that the signaling scheme $\pi^{(U(\pi^*))}$ is exactly the optimal signaling scheme in hindsight π^*. By construction, it is also guaranteed that $\sum_{i\in[m]} \lambda(i)\pi^{(u)}(i) = u$. We note that by focusing on the signaling scheme $\pi^{(u)}$, we bypass the challenge on identifying the value, order, or even signs of $\{\delta(i)\}$. Indeed, for general problem instances, our ConRP does not explicitly learn those quantities, nor can they be inferred from the outcome of ConRP.

Algorithm 2 (ConRP)

Input: number of rounds T, number of states m, prior distribution λ

/* exploring Phase I-identify $\underline{U}$ such that $\underline{U} \le U(\pi^*) \le 2\underline{U}$ */

Initialize $\underline{U} \leftarrow \frac{1}{2}$
while CheckIC($\pi^{(\underline{U})}$) = False **do**
  $\underline{U} \leftarrow \frac{\underline{U}}{2}$
end

/* exploring Phase II-identify a signaling scheme $\pi^\dagger$ such that $U(\pi^\dagger) \ge U(\pi^*) - \frac{1}{T}$ */

Initialize $R \leftarrow 2\underline{U}$, $L \leftarrow \underline{U}$, $\delta \leftarrow 1$
while $R - L \ge \frac{1}{T}$ **do**
  $\varepsilon \leftarrow \frac{\delta}{2}$, $S \leftarrow \lfloor \frac{R-L}{\varepsilon L} \rfloor$, $\ell \leftarrow 1$.
  while CheckIC($\pi^{(L+\ell\varepsilon L)}$) = True **do**
    $R \leftarrow L + \ell\varepsilon L$, $L \leftarrow L + (\ell - 1)\varepsilon L$, $\delta \leftarrow \varepsilon^2$, $\ell \leftarrow \ell + 1$.
  end
end
Set $\pi^\dagger \leftarrow \pi^{(L)}$.

/* exploiting phase */

Use signaling scheme $\pi^\dagger$ for all remaining rounds.

Remark 3. In the ConRP, the first subphase (i.e., exploring Phase I) is used to identify a lower bound and an upper bound of the optimal payoff $U(\pi^*)$ of the platform. This step is crucial for us to establish the $O(\log\log T)$ regret. In Section EC.8 in the Online Appendix, we present simulation studies showing that without this step, the algorithm may perform very badly. Moreover, this subphase is also used to identify an interior point in designing our second main algorithm (Algorithm 5 in the Online Appendix).

We are now ready to describe the main result of this section.

Theorem 1. *The expected regret of* ConRP *is at most* $O(\log\log T)$.

Proof. We analyze the expected regret in exploring Phase I, exploring Phase II, and exploiting each phase separately. We first assume that ConRP finishes exploring Phases I and II before T rounds are exhausted. A similar argument follows for the other case where exploring Phase I or exploring Phase II is completed due to the exhaustion of rounds.

3.1.3. Exploring Phase I. Let $K = -\lceil \log(U(\pi^*)) \rceil$. By definition, CheckIC($\pi^{(2^{-k})}$) = False for $k \in [K-1]$, and CheckIC($\pi^{(2^{-K})}$) = True. Thus, at the end of exploring phase I, $\underline{U}$ is 2^{-K}, and there are K iterations in the

while loop. For each iteration $k \in [K]$, $\texttt{CheckIC}(\pi^{(2^{-k})})$ is called once. By Lemma 4, the total expected regret is

$$\sum_{k\in[K]} \frac{U(\pi^*)}{\sum_{i\in[m]} \lambda(i)\pi^{(2^{-k})}(i)} \overset{(a)}{\leq} \sum_{k\in[K]} \frac{2^{-(K-1)}}{2^{-k}} = \sum_{k\in[K]} 2^{-(K-k-1)} = O(1),$$

where the denominator in the right-hand side of inequality (a) is due to the construction of $\pi^{(2^{-k})}$.

3.1.4. Exploring Phase II. By construction, there are $O(\log\log T)$ iterations in the while loop. Thus, it is sufficient to show the expected regret in each iteration is $O(1)$.

In each iteration k, let $\ell^\dagger \in [S]$ be the smallest index that the signaling scheme $\pi^{(L+\ell^\dagger \varepsilon L)}$ is not IC. The expected regret in iteration k is at most

$$\begin{aligned}
&\sum_{\ell=1}^{\ell^\dagger-1}\left(\frac{U(\pi^*)}{\sum_{i\in[m]}\lambda(i)\pi^{(L+\ell\varepsilon L)}(i)}-1\right)+\frac{U(\pi^*)}{\sum_{i\in[m]}\lambda(i)\pi^{(L+\ell^\dagger\varepsilon L)}(i)}\\
&\overset{(a)}{=}\sum_{\ell=1}^{\ell^\dagger-1}\left(\frac{U(\pi^*)}{L+\ell\varepsilon L}-1\right)+\frac{U(\pi^*)}{L+\ell^\dagger\varepsilon L}\overset{(b)}{\leq}\sum_{\ell=1}^{\ell^\dagger-1}\left(\frac{R}{L}-1\right)+\frac{R}{L}\\
&\overset{(c)}{\leq}(S-1)\frac{R-L}{L}+2\overset{(d)}{\leq}\frac{(R-L)^2}{\varepsilon L^2}+2,
\end{aligned}$$

where equality (a) holds due to the construction of $\pi^{(L+\ell\varepsilon L)}$ and $\pi^{(L+\ell^\dagger\varepsilon L)}$; inequality (b) holds because $U(\pi^*) \leq R$; inequality (c) holds because $\ell^\dagger \leq S$ and $R \leq 2L$; and inequality (d) holds because $S = \lfloor \frac{R-L}{\varepsilon L} \rfloor$.

We finish this part by showing $R - L \leq \sqrt{2\varepsilon}L$ by induction. Let $L^{(k)}, R^{(k)}$, $\delta^{(k)}$, and $\varepsilon^{(k)}$ be the value of L, R, δ, and ε, respectively, in each iteration k. The claim is satisfied for iteration $k = 1$, because $R^{(1)} - L^{(1)} = 2\underline{U} - \underline{U} = L^{(1)}$ and $\varepsilon^{(1)} = 1/2$. Suppose the claim holds for iteration $k-1$. Now, for iteration k, we know that $R^{(k)} - L^{(k)} = \varepsilon^{(k-1)}L^{(k-1)} \leq \varepsilon^{(k-1)}L^{(k)} = \sqrt{\delta^{(k)}}L^{(k)} = \sqrt{2\varepsilon^{(k)}}L^{(k)}$, which finishes the induction.

3.1.5. Exploiting Phase. In this phase, we know that $\pi^\dagger$ is IC and $U(\pi^\dagger) \geq U(\pi^*) - 1/T$, which concludes the proof. □

4. Implementing ConRP for Unknown Ordinal Preference

In this section, we discuss how to adapt the ConRP when the user's ordinal preference, that is, the order of $\{\delta(i)\}$, is not known to the platform. In Section 4.1, we first present the application of ConRP for a general class of user preferences. In Section 4.2, we discuss a more challenging setting where the platform has completely no knowledge about the user's preference. In both scenarios, we show that regret $O(\log\log T)$ is achievable by implementing ConRP.

4.1. Application: Affine State-Dependent User Preference

In this section, we show that, for a general class of users' preferences that are affine with respect to the state, ConRP is able to achieve $O(\log\log T)$ regret, and this regret bound does not depend on the number of possible states.

4.1.1. Affine State-Dependent User Preference. With a slight abuse of the model, in this section, we consider a continuous state space $\Theta \subseteq \mathbb{R}_+$ instead of a finite state space $[m]$. We say a user has an *affine state-dependent preference* if its expected utility $\mathbb{E}_{\theta\sim\mu}[\rho(\theta, a)]$, given posterior $\mu \in \Delta(\Theta)$ and action a, is equal to $\rho(\mathbb{E}_{\theta\sim\mu}[\theta], a)$. The affine state-dependent preference is a fundamental setting in information design literature (Gentzkow and Kamenica 2016, Kolotilin et al. 2017, Kolotilin 2018, Candogan and Strack 2021, Candogan 2022, Arieli et al. 2023).

For a user with an affine state-dependent preference, the user's optimal action $a^* = \arg\max_a \mathbb{E}_{\theta\sim\mu}[\rho(\theta, a)]$ (e.g., whether to watch the video or not), depends only on the expected state $\mathbb{E}_{\theta\sim\mu}[\theta]$ (e.g., expected quality of the video) of the user's posterior belief μ over the underlying states. It is worth highlighting that an affine state-dependent preference could be decomposed as $\rho(\theta, a) = \rho_1(a)\theta + \rho_2(a)$, where functions $\rho_1, \rho_2 : \mathcal{A} \to \mathbb{R}$ are unknown to the platform.

4.1.2. ConRP for Affine State-Dependent Preference. To see how ConRP could be adapted to solve this setting, notice that under affine state-dependent preference, the user's preference difference is essentially $\delta(i) = (\rho_1(1) - \rho_1(0))i + \rho_2(1) - \rho_2(0)$. This shows that there exists at most two possible orders of the user's preference difference $\{\delta(i)\}$, depending on the sign of $\rho_1(1) - \rho_1(0)$. Thus, one can just run ConRP over these two possible orders of $\{\delta(i)\}$ in a round-robin manner and use the payoff of any identified IC signaling scheme to prune out the incorrect order.

The main result in this subsection is summarized as follows.

Proposition 1. *For the affine state-dependent preference, the expected regret of* ConRP *is $O(\log\log T)$; this regret also holds even when the state space Θ is continuous.*

Lastly, we observe that the above results can also be extended to scenarios wherein the user's preferences depend on a potentially nonlinear transformation of the state, denoted as $f(\theta)$, in an arbitrary manner. Fundamental to our conclusions is the observation that

the order of the user's preference difference $\{\delta(i)\}$ is not changing even with such nonlinear transformation.

4.2. Application: User with Unknown Ordinal Preference

In this section, we show that when the order of user's preference differences $\{\delta(i)\}$ is unknown to the platform, a regret $O(m2^{m-1} \cdot \log\log T)$ is achievable by enumerating all possible orders.

Proposition 2. *When the platform has no knowledge of the user's preference, the expected regret of a modified version (see Algorithm* 4 *in Section* EC.4 *of the Online Appendix) of* ConRP *is* $O(m2^{m-1} \cdot \log\log T)$.

Below we briefly discuss how the ConRP could be adapted to obtain the above regret bound. The formal proof of Proposition 2 is provided in Section EC.4 in the Online Appendix. Similar to ConRP, the modified algorithm (Algorithm 4 in the Online Appendix) also relies on a subclass of direct signaling schemes $\pi^{(r,u)}$, but with an additional parameter r to represent a possible (total) order of the user's preference differences $\{\delta(i)\}$. In other words, for every possible order r of the user's preference differences $\{\delta(i)\}$, one can identify a direct signaling scheme $\pi^{(r,u)}$ such that the expected payoff to the platform will exactly equal u if this signaling scheme $\pi^{(r,u)}$ is IC. In the modified algorithm, whenever the algorithm identifies an IC signaling scheme $\pi^{(r,u)}$ (see Lines 4 and 15 in Algorithm 4 in the Online Appendix), it then naturally gives a lower bound of the platform's expected payoff of the optimal signaling scheme, namely, $U(\pi^*) \geq u$. Then one can use the payoff (i.e., u) of such IC signaling scheme to further prune out other signaling schemes that are constructed with different order of $\{\delta(i)\}$ but are either non-IC or have payoff less than u. We would like to note that without pruning, the regret of the algorithm might have linear dependence on the time horizon T.

In more detail, recall that when user's utility function is unknown, the bang-per-buck as well as the true total order r^* are unknown to the platform. In the exploring phase of Algorithm 4 in the Online Appendix, our online policy maintains a subset $\mathcal{P}$ of total orders over $[m]$ that contains the optimal order r^*. In particular, the algorithm initializes the set $\mathcal{P}$ such that it contains all possible orders, and each order r in the set $\mathcal{P}$ specifies a state $i^\dagger \in [m]$, a subset of states that are below the state $i^\dagger$, and a subset of states that are above the state $i^\dagger$. Given an arbitrary order r, we define $\pi^{(r,u)}$ to be the direct signaling scheme as follows: let state $i^\dagger$ be the state associated with this order r, then (a) for every state $i \neq i^\dagger$, $\pi^{(r,u)}(i) = \mathbb{1}[r(i) > r(i^\dagger)]$,[11] and (b) $\pi^{(r,u)}(i^\dagger) = \frac{u - \sum_{i \neq i^\dagger} \lambda(i)\pi^{(r,u)}(i)}{\lambda(i^\dagger)}$. As a sanity check, observe that the signaling scheme $\pi^{(r^*, U(\pi^*))}$ is exactly the optimal signaling scheme in hindsight π^*, and it is also guaranteed that $\sum_{i\in[m]} \lambda(i)\pi^{(r,u)}(i) = u$ by construction.

Note that, although the number of all possible total orders could be as large as $O(m!)$, we can have a more succinct representation on user's ordinal preference. To see this, note that each possible order r can first specify a state $i^\dagger$, a subset of states that are below the state $i^\dagger$, and remaining states that are above the state $i^\dagger$. Then, in total, there are at most $O(m2^{m-2})$ such possible orders.

Remark 4. In both exploring Phases I and II, Algorithm 4 in the Online Appendix checks whether there exists $r \in \mathcal{P}$ such that $\texttt{CheckIC}(\pi^{(r,u)}) = \text{True}$ for some u. Our regret bound has an (2^m) dependence due to brute-force searching over r. We would like to note that (i) in many practical applications, the number of states m is small or even constant, and thus our main focus in this section is the optimal dependence on the number of rounds T, and (ii) when N $(\leq m2^{m-1})$ identical problem instances are allowed to run in parallel, the regret dependence on m becomes $m2^{m-1}/N$.

5. Alternative LP-Based Algorithm

In this section, we provide our second main result: a linear program–based recommendation policy (LP-RP) with $O(\text{poly}(m \log T))$ regret when the user's preferences (including both the cardinal preference and ordinal preference) are unknown to the platform. The main result is as follows.

Theorem 2. *The expected regret of* LP-RP *is at most* $O(m^6 \log^{O(1)}(mT))$.

The proposed LP-RP uses a subroutine MembershipLP: an algorithm (Lee et al. 2018) to solve linear program with membership oracle access.[12] We first formally introduce the linear program optimization with membership oracle access and discuss its connection to our online Bayesian recommendation problem. Then we provide the formal description and the explanation of LP-RP where we also present the proof of Theorem 2.

5.1. Linear Program Optimization with Membership Oracle Access

Optimizing a linear function $f(\cdot)$ within an unknown convex set H has been studied extensively in the literature. There are two standard oracle assumptions: *membership oracle* and *separation oracle*. A membership oracle returns whether a queried point y is contained in convex set H. In contrast, a separation oracle not only returns whether a queried point y is contained in convex set H but also returns a hyperplane that separates y from H if $y \notin H$.

Recall that, in our problem, the optimal signaling scheme in hindsight π^* is the optimal solution of

linear Program ($\mathcal{P}^{\text{opt}}$). From the platform's perspective, the only unknown component in this program is $\{\delta(i)\lambda(i)\}$ in the IC constraint. Nonetheless, using Procedure 1, the platform can determine the incentive compatibility (i.e., whether the IC constraint is satisfied) of any direct signaling scheme. In other words, Procedure 1 works like a membership oracle for the convex set that contains all IC signaling schemes. Thus, finding the optimal signaling scheme π^* can be formulated as optimizing a linear program with membership oracle access. In particular, we leverage the algorithm introduced in Lee et al. (2018) with the following guarantee.

Theorem 3 (Lee et al. 2018). *For any linear function f, and convex set $H \subseteq \mathbb{R}^m$, given an interior point $x^{(0)}$, a lower bound r, an upper bound R such that $\mathbf{B}_2(x^{(0)}, r) \subseteq H \subseteq \mathbf{B}_2(x^{(0)}, R)$,[13] and given a membership oracle, there exists an algorithm* MembershipLP *that finds an ϵ-approximate optimal solution for f in H with probability $1-\delta$, using $O(m^2 \log^{O(1)}(mR/\epsilon\delta r))$ queries to the oracle.*

We note that when only membership oracle is given, the interior point $x^{(0)}$, as well as lower bound r and upper bound R such that $\mathbf{B}_2(x^{(0)}, r) \subseteq H \subseteq \mathbf{B}_2(x^{(0)}, R)$, is necessary for any algorithms. Otherwise, there is an information-theoretic barrier (Grötschel et al. 2012).

5.2. Toward $O(\text{poly}(m \log T))$ Regret

Before we describe our algorithm, let us highlight two major hurdles in applying the membership oracle approach to solve our Bayesian recommendation problem.

1. Although Theorem 3 upper bounds the total number of queries to the membership oracle (a.k.a., Procedure 1), as illustrated in our second example presented in Section 3, the regret from one execution of Procedure 1 may be superconstant.
2. Theorem 3 requires an interior point x as well as a lower bound radius r and an upper bound radius R such that $\mathbf{B}_2(x, r) \subseteq H \subseteq \mathbf{B}_2(x, R)$, and the number of queries depends on the value of r and R. However, in our problem, the interior point is *not* given explicitly. How to find a proper interior point x with nontrivial lower bound radius r (without incurring too much regret) is not obvious in our problem.

5.2.1. Overview of the Algorithm. We now sketch LP-RP. The formal description of LP-RP and its detailed proof is provided in Section EC.5 in the Online Appendix. The high-level idea of this algorithm is to use MembershipLP as a subroutine to identify a signaling scheme $\pi^\dagger$ whose per-round expected regret is $1/T$. In more detail, LP-RP divides the whole T rounds into an *exploring phase* and an *exploiting phase*. In exploring phase, we use MembershipLP to identify a persuasive signaling scheme $\pi^\dagger$, and exploiting phase uses $\pi^\dagger$ until the rounds are exhausted. As mentioned in the above two hurdles, to use MembershipLP, we need to ensure that each query (i.e., a signaling scheme) to MembershipLP cannot incur too much regret, that is, Procedure 1 for checking the persuasiveness of a queried signaling scheme cannot be large. Also, we need to find a proper interior point with nontrivial lower bound radius. To achieve this, there are three subphases in the exploring phase.

Exploring Phase I—Lower Bounding $U(\pi^*)$: Similar to ConRP, the first step of LP-RP is to identify a lower bound and an upper bound of $U(\pi^*)$. But different from ConRP, here, we identify a set $\hat{\mathcal{S}}$ of persuasive direct signaling schemes such that, for every signaling scheme $\pi^l \in \hat{\mathcal{S}}$, it has the following two properties: (i) it has the same payoff $\underline{\mathrm{U}}$ with other signaling schemes in set $\hat{\mathcal{S}}$, that is, $\underline{\mathrm{U}} \equiv U(\pi^l)$, $\forall \pi^l \in \hat{\mathcal{S}}$, and $\underline{\mathrm{U}}$ is relatively good, that is, $\underline{\mathrm{U}} \geq \frac{U(\pi^*)}{m^2}$; and (ii) signaling scheme π^l has a specific structure where it has nonzero probability for recommending action 1 on at most two states.

Lemma 5 (Informal). *When exploring Phase I terminates, $\underline{\mathrm{U}} \geq \frac{U(\pi^*)}{m^2}$ and $\hat{\mathcal{S}}$ is not empty.*

At a high level, Property (i) implies that $\underline{\mathrm{U}} \leq U(\pi^*) \leq m^2 \underline{\mathrm{U}}$, which can guarantee us whenever we use Procedure 1 (as a membership oracle) to check the persuasiveness of a direct signaling scheme in the later rounds, the expected regret is at most $O(m^2)$. Property (ii) can guarantee that we find an interior point with nontrivial lower bound radius r in the later subphase.

Exploring Phase II—Excluding Degenerate States: To find the interior point for Program ($\mathcal{P}^{\text{opt}}$), however, we first note that it is possible the convex set in Program ($\mathcal{P}^{\text{opt}}$) is degenerate and thus no interior point exists. Nonetheless, those degenerate dimensions (i.e., states) must contribute little to $U(\pi^*)$. Thus, in this exploring phase, we use the signaling schemes in $\hat{\mathcal{S}}$ obtained in exploring Phase I to exclude those states and obtain a set $\tilde{\Theta} \subseteq [m]$ that contains all relatively good states (i.e., states whose ω cannot be too negative).

Lemma 6. *When exploring Phase II terminates,*

- *For each state $i \in \tilde{\Theta}$: $\delta(i)\lambda(i) \geq -mT \cdot \max_{j\in[m]} \delta(j)\lambda(j)$;*
- *For each state $i \notin \tilde{\Theta}$: $\delta(i)\lambda(i) < -\frac{mT}{3} \cdot \max_{j\in[m]} \delta(j)\lambda(j)$.*

With the obtained $\tilde{\Theta}$ at hand, we show that there exists a persuasive direct signaling scheme $\tilde{\pi}^{(0)}$ such that $\frac{1}{8m^2T} \leq \tilde{\pi}^{(0)}(i) \leq 1 - \frac{1}{8m^2T}$ for every $i \in \tilde{\Theta}$, and $\tilde{\pi}^{(0)}(i) = 0$

for every $i \notin \tilde{\Theta}$. Furthermore, signaling scheme $\tilde{\pi}^{(0)}$ is an interior point[14] of the following linear program.

$$\begin{aligned} \max_{\pi:\pi(i)=0\ \forall i\notin\tilde{\Theta}} \quad & \sum_{i\in\tilde{\Theta}} \lambda(i)\pi(i) && \text{s.t.} \\ & \sum_{i\in\tilde{\Theta}} \delta(i)\lambda(i)\pi(i) \geq 0 && \\ & \sum_{i\in\tilde{\Theta}} \lambda(i)\pi(i) \geq \frac{1}{16}\underline{U} && \\ & \pi(i) \in [0,1] && i \in \tilde{\Theta} \end{aligned} \qquad (\mathcal{P}^{\mathtt{opt}}_{\tilde{\Theta}})$$

From Lemma 6, we know that the optimal objective value of Program ($\mathcal{P}^{\mathtt{opt}}_{\tilde{\Theta}}$) is close to $U(\pi^*)$.

Lemma 7. *Let $\pi^{\ddagger}$ be the optimal solution in Program ($\mathcal{P}^{\mathtt{opt}}_{\tilde{\Theta}}$), that is, $\pi^{\ddagger} = \arg\max \mathcal{P}^{\mathtt{opt}}_{\tilde{\Theta}}$. Then $U(\pi^{\ddagger}) \geq U(\pi^*) - O(\frac{1}{T})$.*

Exploring Phase III—Executing MembershipLP **with Interior Point Candidates:** In this phase, we identify a direct signaling scheme $\pi^{\dagger}$ whose per-round expected regret is $O(\frac{1}{T})$ (i.e., $U(\pi^{\dagger}) \geq U(\pi^*) - O(\frac{1}{T})$) with probability $1 - \frac{1}{T}$. To do this, we solve a $\frac{1}{T}$-approximate solution in Program ($\mathcal{P}^{\mathtt{opt}}_{\tilde{\Theta}}$) by using MembershipLP as a subroutine. However, we note that we cannot directly identify the interior point $\tilde{\pi}^{(0)}$ to Program ($\mathcal{P}^{\mathtt{opt}}_{\tilde{\Theta}}$ mentioned in exploring Phase II. Instead, we introduce a specific modification for signaling schemes in $\hat{\mathcal{S}}$: For every signaling scheme $\pi^I \in \hat{\mathcal{S}}$, its modification $\pi^{(0)}$ is an interior point candidate. In particular, because of Lemma 6, there exists a signaling scheme $\pi^I \in \hat{\mathcal{S}}$ whose modification $\pi^{(0)}$ is indeed an interior point $\tilde{\pi}^{(0)}$.

Lemma 8 (Informal). *There exists a signaling scheme $\pi^I \in \hat{\mathcal{S}}$ such that its modification $\pi^{(0)}$ is an interior point of Program ($\mathcal{P}^{\mathtt{opt}}_{\tilde{\Theta}}$). In particular, let $r = \frac{1}{16m^2T}$, then $\mathbf{B}_2(\pi^{(0)}, r) \subseteq H(\mathcal{P}^{\mathtt{opt}}_{\tilde{\Theta}})$.*

Finally, we run MembershipLP based on every interior point candidate $\pi^{(0)}$ (and in the end, we pick the best solution as $\pi^{\dagger}$), where we set the interior point $x^{(0)} \leftarrow \pi^{(0)}$, lower-bound radius $r \leftarrow \frac{1}{16m^2T}$, upper-bound radius $R \leftarrow \sqrt{m}$, precision $\epsilon \leftarrow \frac{1}{T}$, and success probability $\delta \leftarrow \frac{1}{T}$.[15]

6. Matching $\Omega(\log\log T)$ Regret Lower Bound

We now show a tight lower bound of $\Omega(\log\log T)$ regret of any online policy, even when the number of states is two and the user has the affine state-dependent preference. Here we allow the online policy to be randomized (i.e., can commit to different signaling schemes at random) and have nonbinary (but finite) signal spaces (i.e., can have multiple recommendation levels).

Theorem 4. *No online policy can achieve an expected regret better than $\Omega(\log\log T)$, even for the family of binary-state problem instances.*

We note that for every binary-state instance, the user's utility can be represented as an affine function over the state space. Thus, the above regret lower bound also holds when the user's utility is an affine function over the state space.

6.1. Overview of the Proof

To show Theorem 4, we focus on problem instances with binary state. Our proof mainly consists of two steps. In the first step, we show that for problem instances with binary state, any online policy can be transformed into a randomized online policy that only uses signaling schemes with binary signal space. This statement is no longer true for general problem instances with nonbinary state, because the classic revelation principle fails. The key technical ingredient (Lemma EC.10 in the Online Appendix) is to show that any posterior distribution of binary state can be induced by a convex combination of signaling schemes with binary signal space, which may be of independent interest. In the second step, we show a reduction from the single-item dynamic pricing problem to our online Bayesian recommendation problem with binary state. Thus, the $\Omega(\log\log T)$ regret lower bound known in dynamic pricing problem (Kleinberg and Leighton 2003) can be extended to our problem. (See more discussions about the connection between our problem and the dynamic pricing problem in Section EC.2 in the Online Appendix.)

Below we provide detailed discussion and related lemmas for the above mentioned two steps. At the end of this section, we combine all pieces together to conclude the proof of Theorem 4.

6.2. Step 1: Binary Signals Suffice

Our first step is to show that every online policy can be transformed into a randomized online policy with binary signal space. Although this might appear obvious at first as binary signal suffices in the optimal signaling scheme in hindsight, it is not a priori clear whether restricting to binary signals is without loss in an online policy without knowing user's utility.

Lemma 9. *Given any problem instance with binary state, for any online policy* ALG, *there exists an online policy* ALG† *that only uses signaling schemes with binary signal space and has regret* $\mathbf{REG}[\mathtt{ALG}^{\dagger}] = \mathbf{REG}[\mathtt{ALG}]$.

We now first sketch the intuition behind Lemma 9. Fix an arbitrary online policy ALG, we construct a randomized online policy ALG† with binary signal space that uses the original policy ALG as a black box. Briefly speaking, in each round t, policy ALG† first asks which

signaling scheme π_t is used by ALG in this round. Then, ALG† uses a signaling scheme $\pi_t^\dagger$ with binary signal space at random[16] such that the distribution of user t's posterior belief induced in $\pi_t^\dagger$ (over the randomness of state, signaling scheme $\pi_t^\dagger$ used by ALG, and $\pi_t^\dagger$ itself) is the same as the one induced by π_t. Note that from user t's perspective, the user's best response is uniquely determined by the user's posterior belief. Thus, the distribution of user t's action is the same in both ALG and ALG†. Finally, ALG† sends user t's action a_t as the feedback to ALG, and moves to the next round. The formal proof of Lemma 9 is in Section EC.6 in the Online Appendix.

6.3. Step 2: Reduction from Dynamic Pricing

The second step in the proof of Theorem 4 is a reduction from the single-item dynamic pricing problem to our online Bayesian recommendation problem. The definition of single-item dynamic pricing problem is as follows.

Definition 3. In the *single-item dynamic pricing problem*, there is a seller with unlimited units of a single item and T buyers. In each round $t \in [T]$, the seller wants to sell a new unit of the item (by setting a price p_t) to buyer t. Buyer t has a private value v^* that is unknown to the seller and will buy the item (and pay p_t) if and only if $v^* \geq p_t$. The regret of a dynamic pricing mechanism ALG is

$$\mathbf{REG}[\texttt{ALG}] \triangleq T \cdot v^* - \mathbb{E}_{p_1,\ldots,p_T}\left[\sum_{t\in[T]} p_t \cdot \mathbb{1}[p_t \leq v^*]\right],$$

where p_t is the price posted by ALG in each round $t \in [T]$.

Theorem 5 (Kleinberg and Leighton 2003). *In the single-item dynamic pricing problem, no randomized dynamic pricing mechanism can achieve an expected regret better than* $\Omega(\log\log T)$.

The following lemma (its formal proof is deferred to Section EC.6 in the Online Appendix) formally states the reduction from the single-item dynamic pricing problem to our online Bayesian recommendation problem.

Lemma 10. *For every single-item dynamic pricing problem instance I, there exists an online Bayesian recommendation problem instance $I^\dagger$ with binary state. For every online policy* ALG† *with binary signal space and regret* $\mathbf{REG}_{I^\dagger}[\texttt{ALG}^\dagger]$ *on online Bayesian recommendation instance $I^\dagger$, there exists a dynamic pricing mechanism* ALG *with regret* $\mathbf{REG}_I[\texttt{ALG}] \leq \mathbf{REG}_{I^\dagger}[\texttt{ALG}^\dagger] + 1$ *on dynamic pricing instance I.*

Remark 5. We would like to note that the binary-state instance constructed in the proof of Lemma 10 also satisfies that the user's utility function is an affine function over the states.

Putting all pieces together, we are ready to prove Theorem 4.

Proof of Theorem 4. Combining Theorem 5 and Lemma 10, in the online Bayesian recommendation problem with binary state, no randomized online policy with binary signal space can achieve an expected regret better than $\Omega(\log\log T)$. Invoking Lemma 9 finishes the proof. □

7. Extensions

In this section, we briefly discuss three extensions: (i) the platform has a state-dependent utility function, (ii) users have an unknown mis-specified prior belief, and (iii) the platform has access to different video categories and must decide which category to display and how. For the first two extensions, the regret bounds for both ConRP and LP-RP remain valid with some algorithm modifications. In the third extension, the double logarithmic regret dependence still holds but is multiplied by the number of video categories. More details and all formal results are in Section EC.7 in the Online Appendix.

7.1. Platform with State-Dependent Utility Function

Recall that our baseline model assumes that the platform's utility function ξ is state independent, that is, $\xi(\theta, a) \equiv a$. We can relax this assumption and consider a more general state-dependent platform's utility function $\xi : [m] \times \mathcal{A} \to \mathbb{R}$. Additionally, we assume that $\xi(i, 0) = 0$ and $\xi(i, 1) \in [0, 1]$ for every state $i \in [m]$. In this more general model, it can be verified that the regret guarantees in Theorem 1 and Theorem 2 continue to hold for modified versions of Algorithm 2 and Algorithm 5 in the Online Appendix. We also note that our state-dependent utility formulation also captures a scenario where the platform is a benevolent player that aims to maximize the social welfare (see Section EC.7 in the Online Appendix for detailed discussions).

7.2. Users with Mis-Specified Beliefs

Our algorithm and results can be extended to the setting where users have mis-specified beliefs for the underlying state realization (Alonso and Camara 2016). In particular, we allow that the users have a mis-specified prior belief $\lambda^\dagger \in \Delta([m])$, which is unknown to the platform, whereas the platform has the prior belief $\lambda \in \Delta([m])$, which may not be necessarily the same as the users' prior belief $\lambda^\dagger$. In this setup, we can show that our Algorithm 2, Algorithm 5 in the Online Appendix, and corresponding regret guarantees continue to hold.

7.3. Which Video to Display

Our results can also be generalized to a setting where there are different categories of videos, and each category could include videos that are in a particular genre or style. For example, sports video can be categorized based on the specific type of sport, for example, soccer, basketball, baseball, American football, and so on, or it can be categorized into subgenres based on the content and purpose of the videos, for example, highlights and clips, training and tutorials, interviews and profiles, and so on. Meanwhile, each category may include videos that have different characteristics (i.e., states). The users not only have different preferences over different states of the videos but also have preferences over different video categories (e.g., a basketball fan may prefer watching videos in the basketball category). The platform initially does not know the user's preferences and can only display one video to the users at each time round to learn the user's preferences.[17] The platform's goal is to maximize cumulative payoff. This involves identifying the video category with the highest payoff and optimizing recommendations within that category to achieve the best results. We show that our `ConRP` algorithm can still guarantee a double logarithmic expected regret of $O(n \log\log T)$, where n represents the number of categories.

8. Conclusions and Future Work

In this paper, we have studied the online Bayesian recommendation problem with featuring a two-sided information asymmetry where the platform knows the payoff-relevant state but does not know the user's preference (and belief), and the user knows the user's preference but is uncertain about the payoff-relevant state. Focusing on policies that minimize the Stackelberg regret, we present two algorithms. The first algorithm is a conservative recommendation policy (`ConRP`). We show that this algorithm can achieve $O(\log\log T)$ regret when the platform knows the user's ordinal preference over the states. Moreover, this algorithm can also be readily adapted to the setting with unknown ordinal preference. In particular, the same regret $O(\log\log T)$ can be achieved when the user's preference is affine with respect to the state, and regret $O(m2^{m-1} \cdot \log\log T)$ can be achieved for arbitrary preference. Our second algorithm is a linear programming-based algorithm (`LP-RP`) that utilizes the problem structure and can achieve $O(\mathsf{poly}(m \log T))$ regret, which is more desired when the number of states m is large and the user's preference is arbitrary. Our research opens a number of interesting and challenging questions for future research.

8.1. Better Regret Dependency on State Space for Unknown Ordinal Preference?

Our lower bound only establishes regret dependency on the time horizon T; however, it does not rule out the possibility to design an algorithm with achieving $O(\mathsf{poly}(m) \cdot \log\log T)$ regret for unknown ordinal preference setting. Thus, for this setting, it would be interesting to explore whether one can tighten up the lower bound or design an algorithm whose regret has only polynomial dependency on the number of states m with still double-logarithmically depending on time horizon T. Making progress in this direction likely requires a more judicious characterization of the underlying geometry of our online problem.

8.2. Extension with Users' Heterogeneity

In the paper, we focus on the setting where the platform is interacting with users that have the same preference and the same belief. Yet, in some applications, users may be heterogeneous and have different preferences or beliefs over the payoff-relevant states. Thus, an important extension of our problem with significant practical implications would be to consider a setting that captures the users' heterogeneity. In particular, one actionable extension to capture users' heterogeneity is to associate each incoming user at time t with a context $\boldsymbol{x}_t \in \mathcal{X} \subseteq \mathbb{R}^d$ (observed by the platform) where $\mathcal{X}$ is the context space. The utility function of the user with the context $\boldsymbol{x}$ is given by $\rho : [m] \times \mathcal{A} \times \mathcal{X} \to \mathbb{R}$. For example, this utility function could be linear in the context, namely, $\rho(i, a, \boldsymbol{x}_t) = \boldsymbol{\omega}_{i,a}^\top \boldsymbol{x}_t$, where $\{\boldsymbol{\omega}_{i,a}\}_{i\in[m], a\in\mathcal{A}}$ with each $\boldsymbol{\omega}_{i,a} \in \mathbb{R}^d$ are unknown vectors associated with each state action pair that the platform needs to learn through the interactions with the users. With this formulation, our problem then connects to the well-studied contextual dynamic pricing literature. In this line of literature without market noise (Leme and Schneider 2018, Lobel et al. 2018, Cohen et al. 2020, Liu et al. 2021), the unknown valuation v of the buyer for the product is oftentimes assumed to be a linear function of the context $\boldsymbol{x}$ (including product and customer features/characteristics), namely, $v = \boldsymbol{\omega}^\top \boldsymbol{x}$ with some unknown vector $\boldsymbol{\omega}$. Notably, the pricing algorithm developed for this problem is a nontrivial geometric extension of the conservative binary search proposed in Kleinberg and Leighton (2003). Building on the algorithmic insights we developed in Section 3, we believe that it is possible to develop a counterpart contextual algorithm in our setting. Given that this may require substantial efforts and is also beyond the scope of this work, we leave this practically important but also technically interesting problem as future work.

8.3. Generalized Setting with Multiple Signals

In our designed algorithms (see Algorithm 2), the signaling scheme at each time round only uses at most two signals. This is largely motivated by the fact that (i) the users have binary action, (ii) and we consider a

population of homogeneous users who share the same utilities and beliefs. With these facts, the two signals suffice for the offline problem when the platform knew the user's preference and belief. Yet, in practice, we may often observe that the platform uses more than two signals with more involved signaling structure (e.g., YouTube usually displays a rank or order list of videos to the users). The potential reasons behind this observation could be that (i) the users may usually follow a choice model that they have multiple actions that they can take, or (ii) each user may have a private type, and the platform may need to design the signaling scheme without observing the user type. It can be shown that either one of these reasons can lead to the optimal signaling scheme with possibly multiple signals and more involved signaling structure. For example, the offline counterpart of the setting with private types corresponds to a public persuasion problem where the platform designs a public signaling scheme for all users with different types (Dughmi and Xu 2017, Candogan 2022), and the online problem of this setting requires learning the users' type distribution. Exploring the problem that requires general signaling structure may need a different set of algorithmic ideas; thus, we leave this as a direction for future work.

Endnotes

[1] In practice, platforms make joint decisions on which video to display and how to recommend. This work decouples them and focuses on the recommendation problem by assuming that the display decision is made exogenously. In Section 7, we discuss how our results could be generalized if the platform can also choose which video to display.

[2] The Stackelberg regret is a regret notion recently developed for strategic settings (Dong et al. 2018, Chen et al. 2020), which compares to the optimal policy in hindsight, assuming users will correspondingly adapt their behaviors to the benchmark policy (thus the "Stackelberg" in its name). Although previous works demonstrated the difficulty of obtaining sublinear Stackelberg regret in online classification problems (Chen et al. 2020), we surprisingly show that our problem admits efficient online learning algorithms with Stackelberg regret that only has logarithmic dependence on the number of rounds T.

[3] The situations with unknown different user priors are mathematically equivalent, and their results are provided in Section 7.

[4] In practice, the online platform may infer the user's ordinal preference—that is, the relative ranking of videos based on the perceived interest or relevance from offline data like past interactions. However, the platform may find it hard to learn the degree to which a user enjoys a particular video—that is, how much the user prefers watching this video than not watching the video.

[5] In our recommendation problem, two videos should be considered as having different states if (a) the platform has enough information to distinguish them, and (b) the user's utility for watching them is different.

[6] We refer the reader to the work by Dworczak and Pavan (2022) and Babichenko et al. (2021) for a comprehensive overview on different methods in the robust approach.

[7] For ease of presentation, we restrict the signal space to be finite. For continuous signal space Σ, $\pi(i,\cdot)$ can be interpreted as a probability density function (conditional on state i) over the continuous signal space Σ. We note that this restriction is without loss because the optimal signaling scheme characterized in Lemma 1 indeed has finite signal space.

[8] For example, in the literature of Bayesian social learning with pricing, the customers are usually assumed to be able to update their beliefs in a Bayesian manner upon seeing certain new information about the product quality (Ifrach et al. 2019, Shin et al. 2023) and then make the purchase decision by maximizing the corresponding expected utility (Ifrach et al. 2019). Other literature includes signaling in queues (Debo et al. 2012, Lingenbrink and Iyer 2019). We believe that the inclusion of Bayesian rational users inherently adds complexity to the problem, yielding rich insights and results. We view incorporating realistic and relevant behavioral biases as a next step in this research direction.

[9] Throughout this paper, we say a state i is above (respectively, below) state j if it satisfies that $\delta(i) \geq \delta(j)$ (respectively, $\delta(i) < \delta(j)$).

[10] Under this signaling scheme π, if π is IC, then we have $U(\pi^*) \geq U(\pi) \geq C$; otherwise, $U(\pi^*) < U(\pi) < C$.

[11] Given an order r, we denote $r(i)$ by the rank of state i.

[12] `LP-RP` uses MembershipLP as a black box. Namely, it can be replaced by other algorithms for linear program with membership oracle access.

[13] Notation $\mathbf{B}_2(x^{(0)}, r)$ represents the ball of radius r centered at $x^{(0)}$.

[14] Here we mean $\tilde{\pi}^{(0)}$ is an interior point of convex set in Program ($\mathcal{P}^{\mathrm{opt}}_{\tilde{\Theta}}$) when we restrict to states in $\tilde{\Theta}$.

[15] When an incorrect interior point is given, MembershipLP terminates with a suboptimal solution. The number of queries to the oracle is the same as the one in Theorem 3.

[16] Namely, $\mathtt{ALG}^{\dagger}$ randomly picks a signaling scheme $\pi_t^{\dagger}$ and commits to it in round t.

[17] In practice, the platform may display an assortment of videos to the user, with the user's choice of which video to watch potentially following a particular choice model. However, to focus on how the platform can leverage its information advantage about the video state to persuade the user to watch the video, we simplify the scenario by abstracting away from the assortment modeling. Instead, we assume that the platform can only display one video at a time.

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